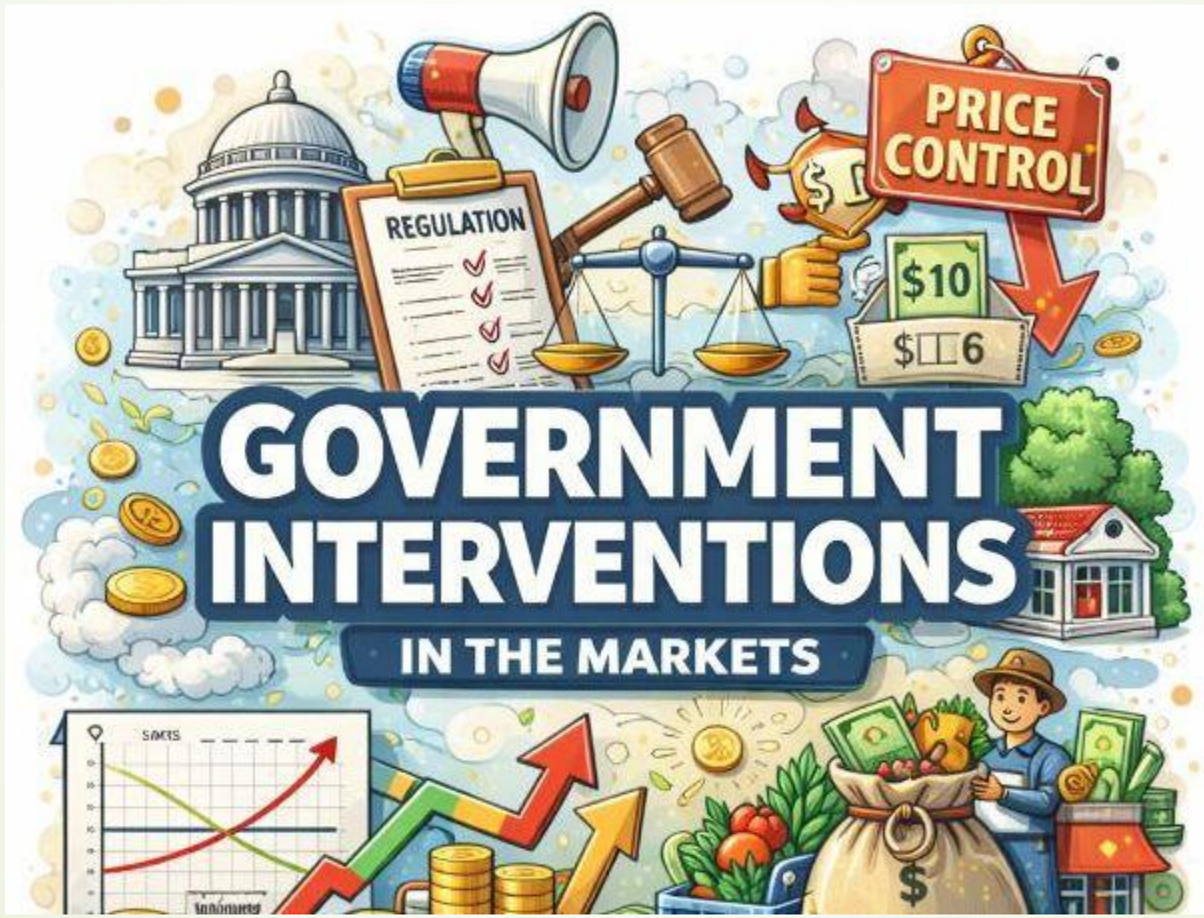
Fundamentals of Business Economics
IS 2211

Dr.R.L.C.Shyama, PhD

Department of IDS

Faculty of IT

Lecture 5
Government Interventions in Markets



Learning Outcomes

At the end of the session, you should be able to:

- Explain why governments intervene in markets
- Identify types of intervention: price controls, taxes, subsidies
- Analyze impacts on price, quantity, and welfare
- Explain the impact of such government interventions

1. Introduction

👉 *"If markets are efficient, why do we still see problems like market failures, price distortions, and inequitable access to goods and services persist?""*

In theory, **free markets** allocate resources efficiently through demand and supply. But in reality, markets can:

•Overproduce harmful goods

•Underproduce beneficial goods

•Create inequality

•Fail to provide essential services

👉 This is where **government intervention** becomes necessary.

2. Why Governments Intervene?

- Government intervention is mainly justified when markets fail to produce efficient or fair outcomes

- **Market failure** occurs when the allocation of resources by the market is not optimal from society's perspective.

👉 Allocative efficiency = resources go to their *best possible use*

- **Government intervention aims** to correct market failures, stabilize economies, ensure fair income distribution, and protect consumers.

2.1 Types of Market Failures

- ✓ Externalities
- ✓ Public goods
- ✓ Information asymmetry
- ✓ Market power (monopoly)
- ✓ Equity issues

Externalities

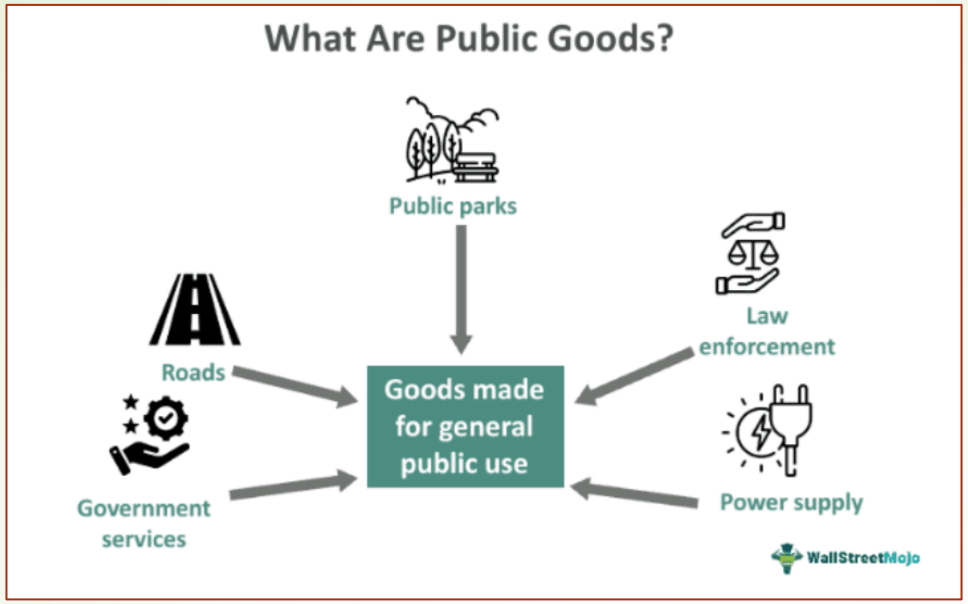
Externalities occur when the actions of consumers or producers affect third parties who are not part of the transaction.

Type	Definition	Example(IT / Digital)	Market Outcome	Economic Problem	Government Response
Negative Externalities	A cost imposed on third parties	Data centers causing pollution; excessive digital advertising affecting users	Overproduction	Social cost > private cost	Taxes, regulation, emission limits
Positive Externalities	A benefit received by third parties	Open-source software; digital education platforms; AI research	Underproduction	Social benefit > private benefit	Subsidies, grants, public support

- **Negative externality:**
    - Firms produce too much because they only consider **private costs**.
    - Example: Cloud computing firms may ignore energy and environmental costs.

- **Positive externality:**
    - Firms underproduce because they only consider **private benefits**.
    - Example: A company developing AI for medical research may not receive full societal benefits, so innovation is limited.

Public Goods



Some goods cannot be efficiently provided by private markets due to their unique characteristics.

Key Characteristics:
•**Non-rival:** One person's consumption does not reduce availability for others.
•**Non-excludable:** Cannot prevent anyone from using the good.

Examples (Digital context):
•Public Wi-Fi infrastructure
•Cybersecurity frameworks
•Open-source development platforms

Problem: Free rider problem → private firms avoid supplying
Government Solution: Direct provision, subsidies, or grants

Information Asymmetry

- *This an economic phenomenon where one party in a transaction possesses more or better knowledge than the other, creating a power imbalance.*

- *This disparity frequently leads to market inefficiencies, such as adverse selection (hidden information) or moral hazard (hidden actions), often causing the less-informed party to make suboptimal decisions*

Examples (Digital / IT):
•Software subscriptions with hidden fees
•Cloud service contracts with complex terms
•Online marketplaces where sellers know product quality better than buyers

Consequences:
•Consumers make poor choices
•Firms may exploit users
•Trust in markets decreases

Government Solution:
•Regulations mandating disclosure
•Digital consumer protection laws
•Standardized labeling of digital services

Market Power (Monopoly / Oligopoly)

Some firms can dominate markets and influence prices or output, creating inefficiencies.

Examples (Digital / IT):

•Major tech platforms controlling app distribution

•Search engine monopolies controlling data access

Consequences:

•High prices

•Reduced choice for consumers

•Limited innovation

Government Response:

•Anti-monopoly laws

•Regulatory oversight

•Encouraging competition through startup support

Equity / Fairness Considerations

Even if the market is efficient, the outcomes may not be socially acceptable.

Examples:

•Unequal access to online education

•Disparities in internet availability across regions

•Digital divide affecting opportunities

Government Role:

•Subsidized digital education

•Grants for rural internet access

•Social welfare programs for technology access

3. Types of Government Intervention

Introduction to Policy Tools

Once market failure is identified, governments use different policy tools to influence market outcomes and improve efficiency or fairness.

Government intervention is not random — each tool is designed to address a **specific type of problem**:

Problem	**Policy Tool**
Overproduction (negative externality)	Tax
Underproduction (positive externality)	Subsidy
High prices (affordability issue)	Price ceiling
Low producer income	Price floor
Unsafe or unfair practices	Regulation
Excess imports / supply	Quotas

1. Price controls: ceilings and floors
2. Taxes: reduce consumption, raise revenue
3. Subsidies: encourage production/consumption
4. Regulation: standards, safety, environment
5. Public provision: goods/services supplied by government

3.1 - Price Controls

Refers to the setting of a **maximal/minimal price above/below** the **equilibrium price** by the **government**.

1.**Governments** set **price controls** to not allow **markets** to reach their **natural equilibrium.**
2.By setting **price controls**, governments force **disequilibrium** in the market, thus creating:

1. **Shortages (excess demand)**
2. **Surpluses (excess supply)**

Price Ceiling (Maximum Price)

A **maximum price**, **below the equilibrium price**, set by the government for a particular good or service.

1. The imposition of a price ceiling ensures that the price at which sellers are allowed to sell a good cannot exceed the maximum price set by the ceiling.

2. This government-imposed limit prevents prices from rising above a specific level for the good or service in question.

- Set below equilibrium price to make goods affordable
- Creates shortage: Qd > Qs
- Non-price rationing: queues, favoritism
- Black markets may emerge
- Long run: reduced supply, lower quality

Real Life Examples

Agreement On Maximum Retail Price of Eggs

by Staff Writer
31-10-2024 | 7:48 AM

COLOMBO (News 1st); The Egg Producers' Association and wholesale traders have agreed to maintain the maximum retail price of eggs at Rs. 37.

Sri Lanka to slap maximum retail price for fuel

Deccan Herald, Newspapers, Oil & Gas July 24, 2023 Binu Mathew

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The Sri Lankan government would introduce from next month a maximum retail price for fuel, State Minister of Energy DV Chanaka said on Sunday.

Speaking to reporters, Chanaka said that the Maximum Retail Price (MRP) would be set as per the pricing formula of the government.

"The government would introduce a maximum retail price for fuel. Based on the pricing formula we will set a MRP," Chanaka said.

The move of the Sri Lankan government comes even as Chinese energy company Sinopec is set to enter the retail fuel market for the first time in August.

READ MORE

Hotline for complaints on vendors selling rice above MRP

October 31, 2023 10:47 am

Follow 491K people are following this.

The Consumer Affairs Authority (CAA) h
selling rice at higher rates beyond the m
are hoarding the stocks.

As such, the complaints can be submitt

Meanwhile, the Authority has initiated ra

A senior official of the CAA pointed out t
Red nor White Nadu Rice can be sold a
Samba is set at Rs. 260, Samba Rice a

In the meantime, there are reports of a
unavailable at most of the supermarkets

Sri Lanka- Maximum retail price imposed on turmeric powder

Date
4/21/2020 11:05:29 AM

Share on Facebook Tweet on Twitter G+ in

(MENAFN - Colombo Gazette) A Maximum Retail Price (MRP) has been imposed on turmeric powder.

The Government announced that an MRP of Rs. 750 per kg has been imposed on turmeric powder from today.

Acknowledgement : Ms. Shalini Upeshka

3. 1 : Price Controls ctd.

Price Ceiling (Maximum Price)

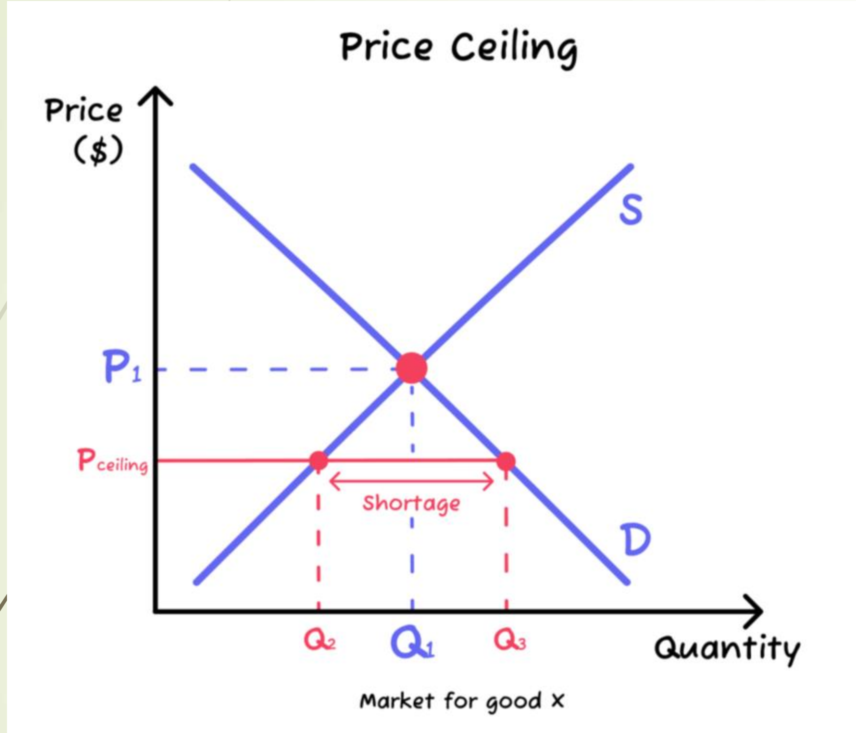
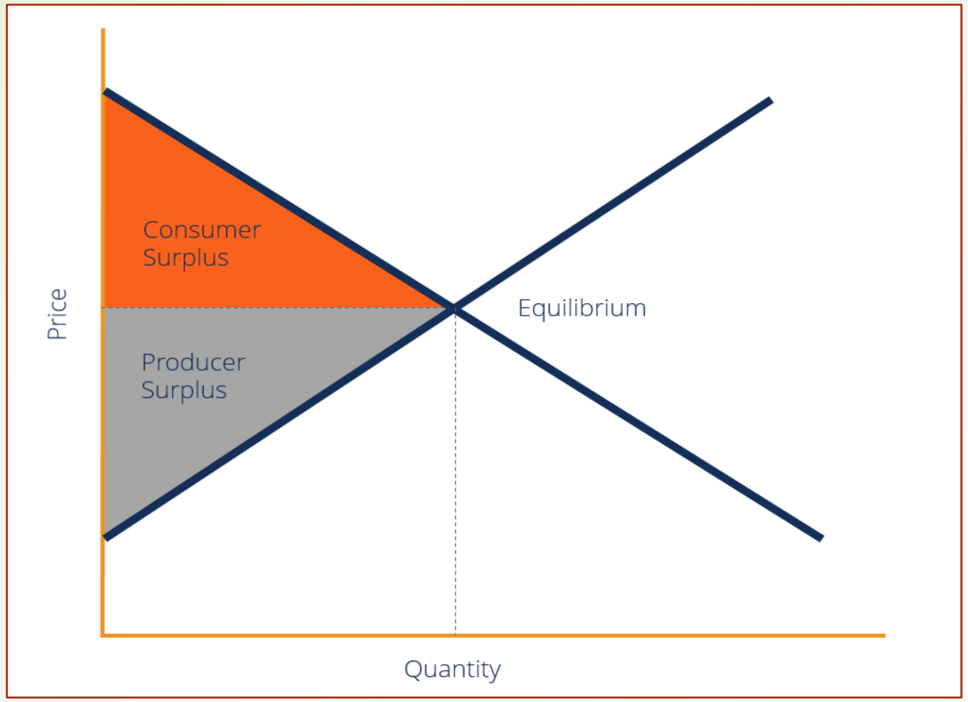


Figure : 1

As seen in Figure 1, at the initial **equilibrium price** (P_1) there is a quantity of $\mathbf{Q_1}$ bread demanded. After the **price ceiling ($P_{ceiling}$)** is imposed:

1. The **quantity demanded increases** to $\mathbf{Q_3}$, while the q**uantity supplied decreas.**
2. Meaning that the **market demand is higher than the market supply**, hence a **shortage of Q_3-Q_2.**

Consumer Surplus & Producer Surplus



- **Consumer surplus**

Consumer surplus is the economic benefit gained by consumers when they purchase a product for less than the maximum price, they were willing to pay. It represents the difference between the total willingness to pay (reservation price) and the actual market price, visually represented as the area below the demand curve and above the equilibrium price

- **Producer surplus**

Producer surplus is the economic benefit producers receive when they sell a product at a market price higher than the minimum price they were willing to accept, which is defined by their marginal cost. Visually represented as the area above the supply curve and below the market price, it measures producer welfare

3.1 : Price Controls ctd. : Welfare loss

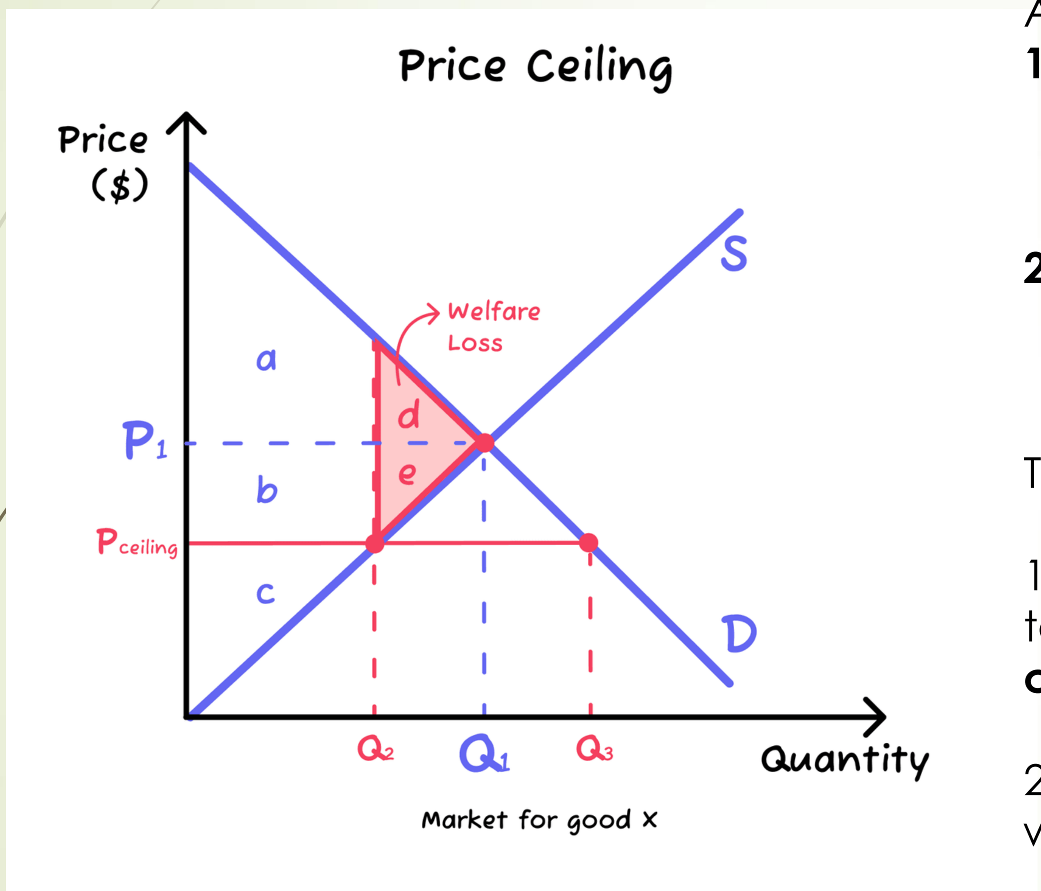


Figure : 2

As it can be seen in *Figure 2*:

1.Before the price ceiling:

1. **Consumer surplus** = a+d
2. **Producer surplus** = b+c+e

2.After the price ceiling(below equilibrium)

1. **Consumer surplus =** a+b
2. **Producer surplus** = c

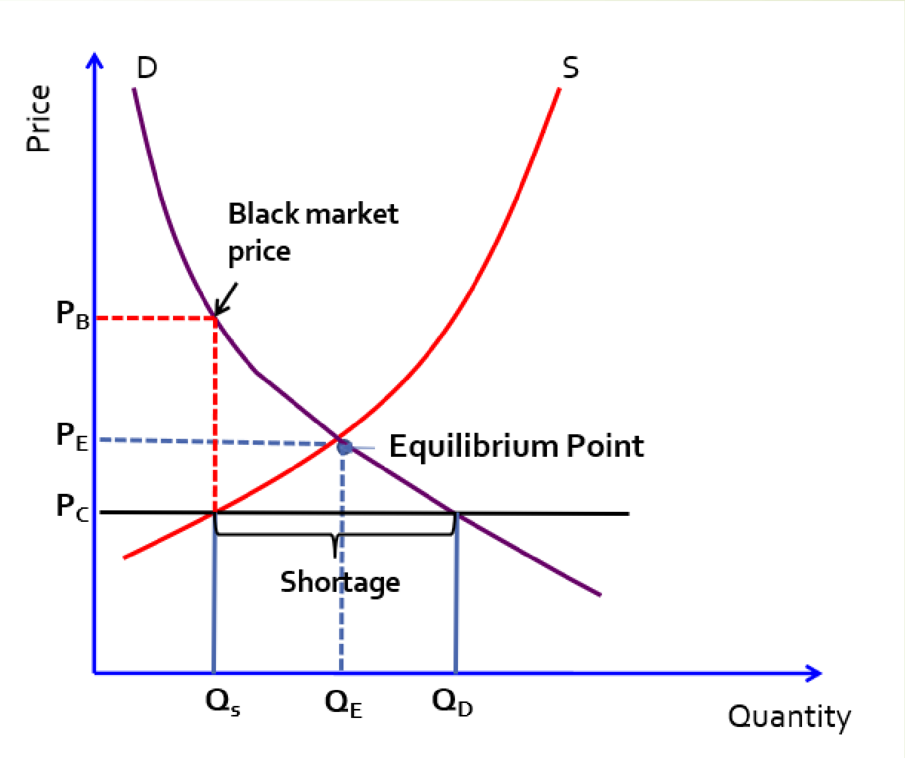
Therefore, there is a **welfare loss (WL)** of the **areas d+e.**

1.This is the amount of **social welfare (surplus)** lost due to **resource misallocation** arising from the **price ceiling.**

2.Remember **allocative efficiency** is achieved when **Marginal benefits (MB) = Marginal Costs (MC).**

How to Manage Consequences of Price Ceiling

- Excess demand (shortage of supply) equal to Q_D-Q_S quantity. Thus, there will be long queues.
- Some suppliers will sell products in black market (illegally) at price P_B.
- Hence, government must introduce rationing schemes. Rationing is the controlled distribution of scarce resources, goods, services, or an artificial restriction of demand.
- Should carefully monitor for black-market operations.



Activity 01

- The below diagram shows the demand and supply for dhal.

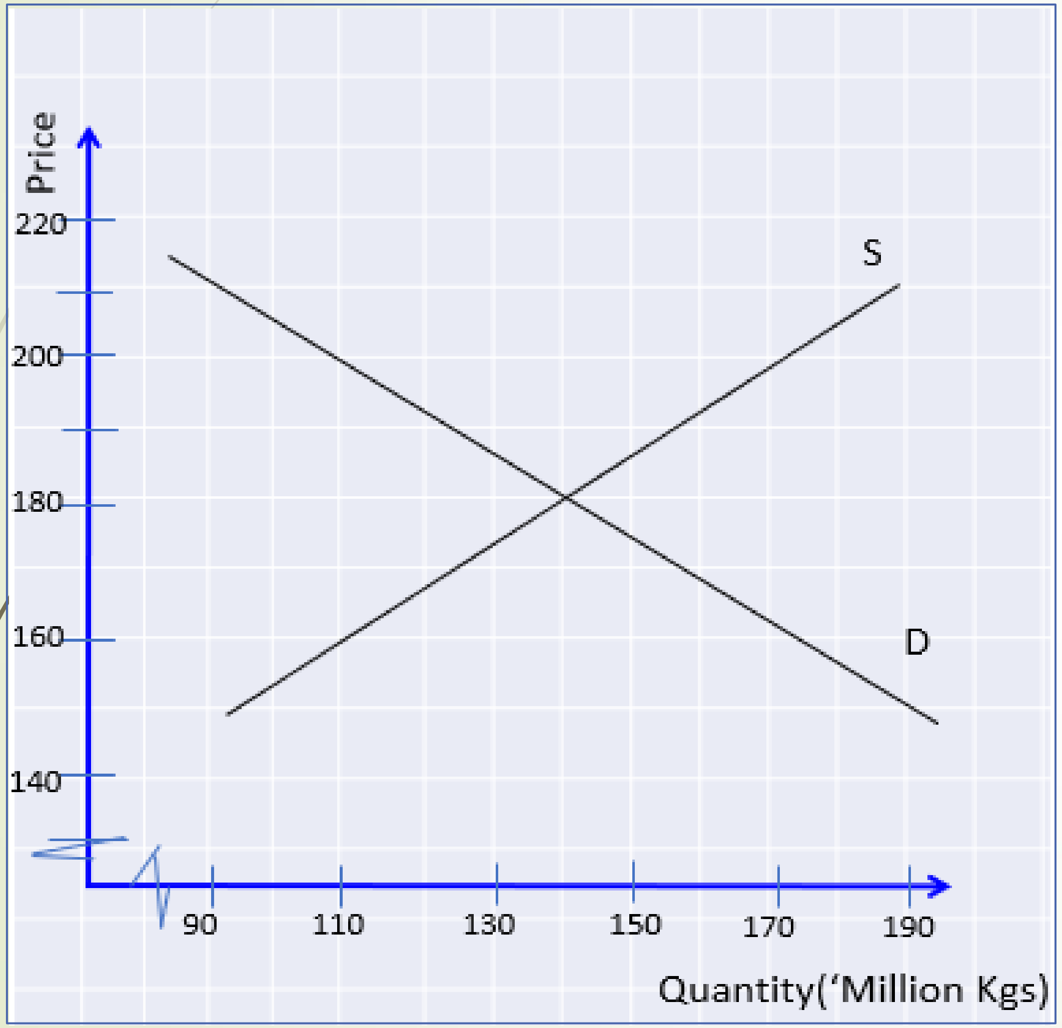


Figure : 3

a) What is the equilibrium market price and quantity?

b) Suppose later the government imposed a maximum price (price ceiling) of Rs. 160.00 per kg. Illustrate the effect in the diagram.

c) What is the government's main intention of setting a price ceiling?

d) Calculate the excess demand as a result of
this price floor.

e) How price floors cause inefficiency in the long run?

Price floors (minimum prices)

- A **minimum price** set by the government for a good, **above the equilibrium** price**.**
- The imposition of a **price floor** ensures that the **price** at which **sellers** are allowed to sell a good **cannot fall below** the **minimum price** set by the **floor**.
- This **government**-imposed limit prevents prices from dropping below a specific level for the good or service in question.

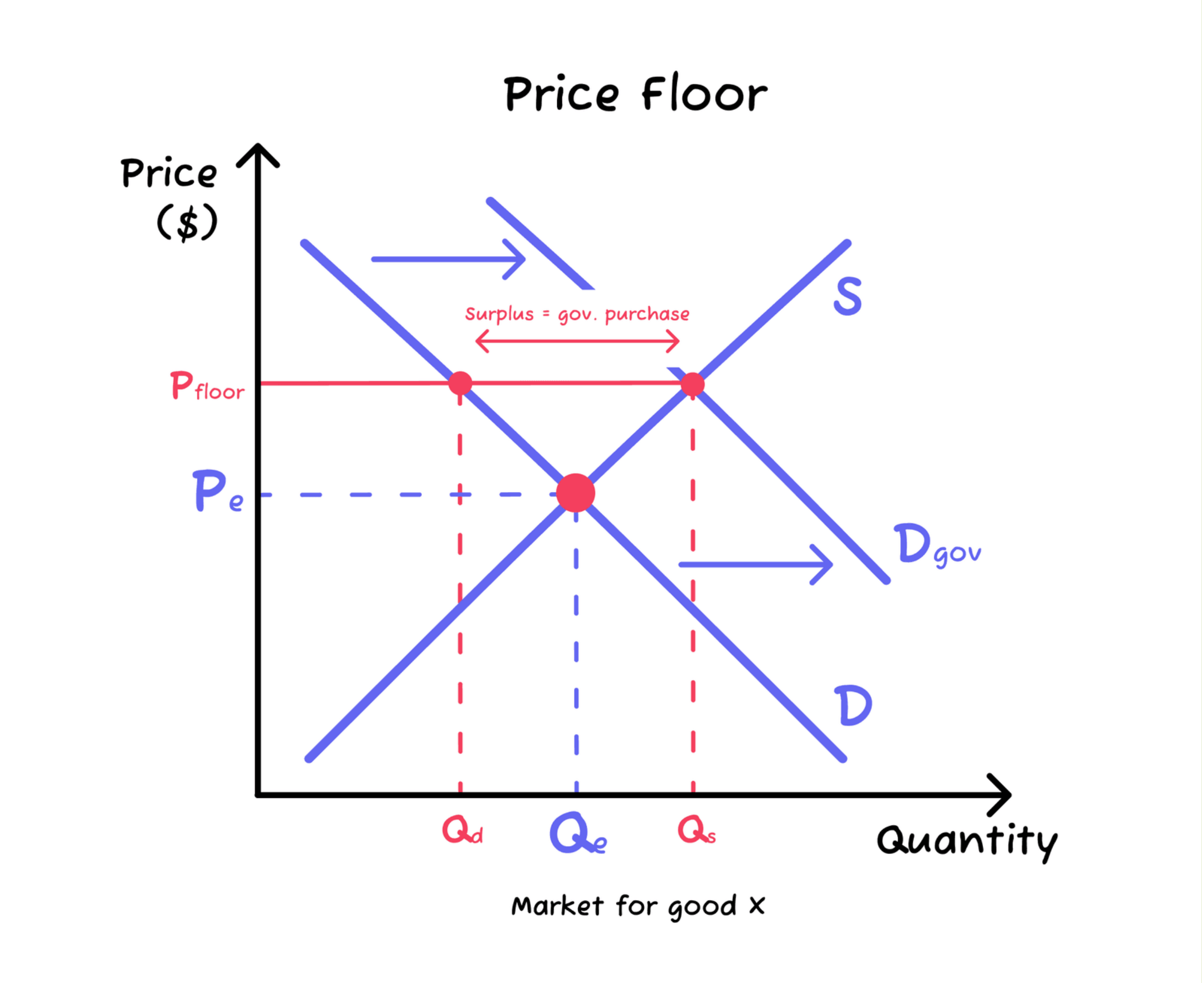


Figure :4

Price floors (minimum prices)

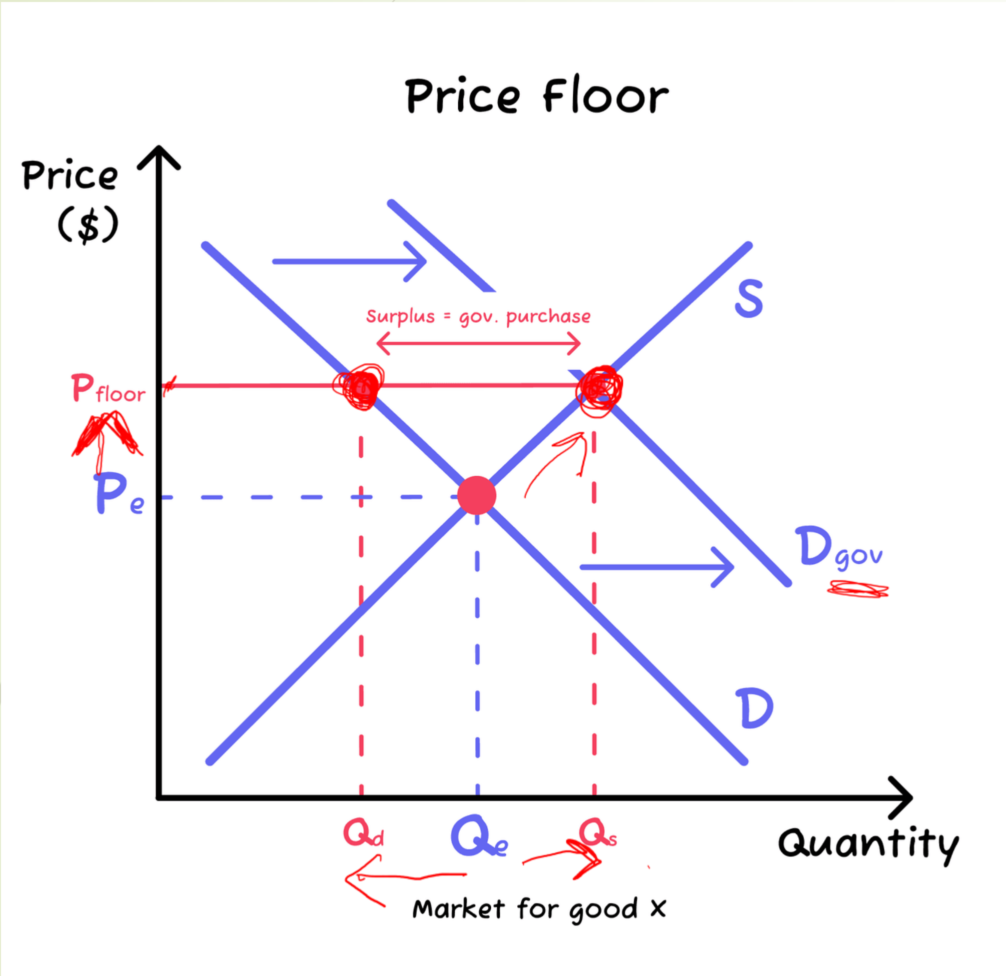


Figure : 4

As it can be seen in *Figure 4*, at the initial **equilibrium price** (**P_e** there is a **quantity Qe** being **demanded**.

1. After the **price floor (P_{floor})** is imposed:
    1. The **quantity demanded decreases** to **Qd**, while the **quantity supplied increases** to **Qs.**
    2. This creates a situation where the **market supply is higher than the market demand**, hence a **surplus of Q_s–Q_d.**
2. This

spare **surplus** that **consumers** are **not willing and able** to **buy** is then **bought by the government**. Then, the **government** either:

1. **Stores the surplus.**
2. **Exports the surplus.**

1. When the government buys the excess supply, the **demand** for the good/service increases, **shifting the demand curve to the right (D→D_{gov}).**

$Q_s > Q_s$ –

Real Life Examples

Minimum room rate is back to city hotels to boost tourism earnings

Monday, 26 June 2023 04:32 - 3969

Tourists enjoy a visit to the Independence Square at Colombo 7 – Pic by Pradeep Pathirana

- **From 1 August five stars required to charge $ 130 plus taxes as opposed to $ 60 at present due to intense price undercutting**
- **The Hotels Association of Sri Lanka President M. Shanthikumar and Colombo City Tourist Hotels Association President Rohan Karr hail impending move; thank Govt. for timely intervention**
- **Express confidence move will help improve financial viability of city hotels; save jobs and boost industry earnings**
- **Dismiss claims MRR will discourage biggest tourist source market India; points to SL not having tapped the fuller potential of Indian market yet**
- **Opines move will help erase bad impression of Colombo being a cheap destination; calls for industry to think out of the box and rally to showcase best of city and Sri Lanka**
- **Colombo has 7,000 rooms and further 2,000 more will open in next 24 months**

Minimum Wage – National Minimum Wage

The current minimum wage in Sri Lanka is LKR12,500.00 per month in 2021. It became valid on August 16, 2021.

- Valid in February 2024
- Minimum wage with effect from August 16, 2021.
- The amounts are in Sri Lanka Rupee (Rs).

JOB TYPE

		th
Workers in any industry or service	LKR	00

Big Onions Sri Lanka will get minimum guaranteed price

The Sri Lankan Finance Ministry has increased the minimum guaranteed price of the government for big onions from Rs 60 to Rs 80 per kilo. According to a statement by the Ministry, big onion farmers were unable to get a fair price in the recent past, due to the drop in large onion prices.

"Therefore, the big onion yields declined and a large amount of foreign exchange was spent on importing big onions," it said.

Considering these facts, the proposal made by Prime Minister Mahinda Rajapaksa in his capacity as the Minister of Finance to increase the Government's guaranteed price for big onions and encourage farmers received the cabinet approval on February 27.

Also, a new program will be launched to increase the cultivation of big onion to be implemented jointly by the Ministry of Mahaweli, ... l Development together with the ... and Policy Development. The progress ... d by the Cabinet of Ministers every six

Govt announce Minimum Purchase Price for paddy varieties

Feb 5, 2025 | 111 | 0

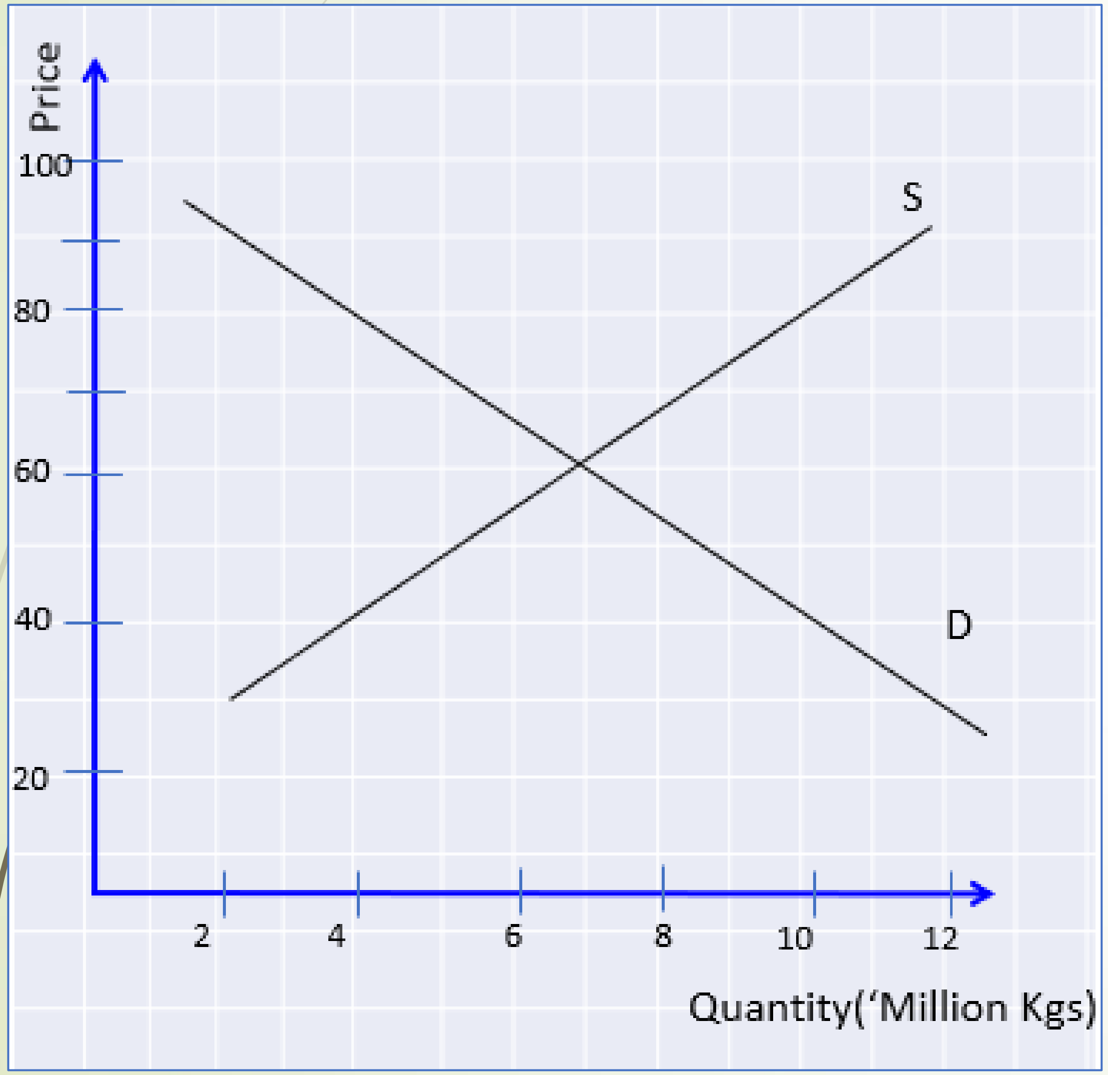
The government announced a Minimum Purchasing Price (MPP) for paddy varieties that would come into effect from tomorrow onwards, Agriculture Minister K.D Lal Kantha announced today.

Accordingly, the Paddy Marketing Board will purchase paddy at the following rates per kilogram: Rs.120 for Dried Nadu rice, Rs.125 for Samba, and Rs.132 for Keeri Samba.

The government came under severe criticism in recent weeks for delaying to announce MPP for paddy as farmers alleged that they were forced to sell their harvest at lower rate by rice millers.

Activity 02

- The below diagram shows the demand and supply for big onions.



a) What is the equilibrium market price and quantity?

b) Suppose later the government increased the minimum price to Rs. 80.00 per kg (price floor). Illustrate the effect in the diagram.

c) What is the government's main intention of setting a price floor?

d) Calculate the excess supply as a result of this price floor.

e) How price floors cause inefficiency in the long run?

Do Price Ceilings and Floors Change Demand or Supply?

- Price ceilings and price floors do not change demand or supply, as they do not shift the demand or supply curves.
- Instead, they impose a legal price restriction that prevents the market from reaching its equilibrium price.
- As a result, they cause a movement along the demand and supply curves, leading to changes in quantity demanded and quantity supplied.
- The original equilibrium remains unchanged, but the market operates at a disequilibrium outcome (shortage or surplus).

3.2 -Taxes and Subsidies

Governments use **taxes and subsidies** as key tools to influence how markets operate. These policies are especially important when markets fail to produce efficient or socially desirable outcomes.

- Taxes → discourage production or consumption
- Subsidies → encourage production or consumption
- Both affect **price, quantity, and welfare**

👉 Central idea: Government changes **market incentives**

3.2 -Taxes

A **tax** is a compulsory payment imposed by the government on goods, services, or income.

From an economic perspective:

✓ A tax increases the **cost per unit** of a good

✓ This directly affects firms' production decisions and consumers' purchasing decisions

👉 Important:

Even if the tax is imposed on producers, the **entire market adjusts**, so both consumers and producers are affected.

DIRECT TAXES	INDIRECT TAXES
These are paid directly by the individual or entity to the central/state government	These are paid by one individual/ entity to the government and subsequently the tax burden is passed on to a different entity/individual

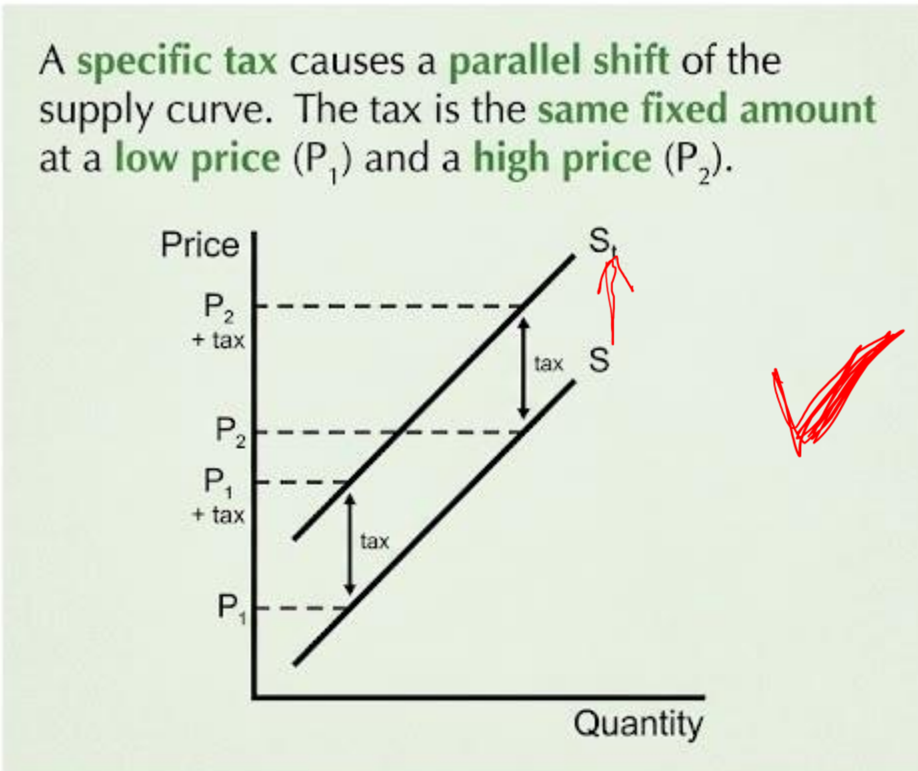
Direct Tax	**Indirect Tax**
Personal Income Tax (PAYE) – tax on salaries	Value Added Tax (VAT) – charged on goods & services
Corporate Income Tax – tax on company profits	Excise Duty – on alcohol, cigarettes, vehicles
Withholding Tax (WHT) – on interest, dividends	Customs Duty – on imported goods
Capital Gains Tax – on sale of property/land	VAT on restaurant bills and supermarket purchases
Tax paid directly by individuals or firms	Tax paid indirectly through prices
Burden cannot be shifted to others	Burden can be shifted to consumers

Taxes and Businesses

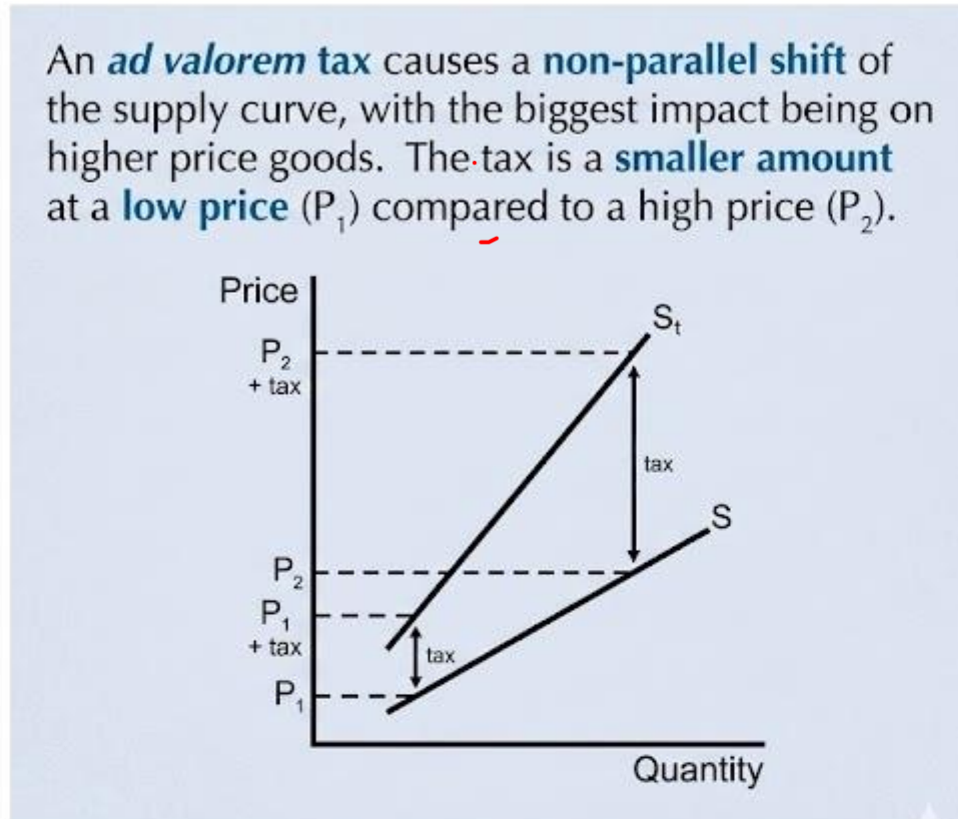
- Businesses are responsible for paying taxes on profits, on capital gains on the sales of assets,
  while many charities have certain tax exemptions which they can claim.
- Many businesses, large and small, will employ accountants with the expertise in the tax system to help them comply with the law and avoid paying taxes when not necessary.
- **Tax avoidance**: The legal use of the tax system to minimize the amount of tax paid.
- **Tax evasion**: The deliberate and illegal activity by individual or organization not to pay the required statutory tax due.
- Tax avoidance is not illegal, it is using the tax system to benefit the business whereas tax evasion is illegal as it is the deliberate non-payment of tax due. While many large firms can legally avoid tax, the morality of doing so has been questioned.

How taxes on Sellers affect Market Outcomes

- When it comes to tax on sellers, the tax can be specific taxes and Ad Valorem taxes.



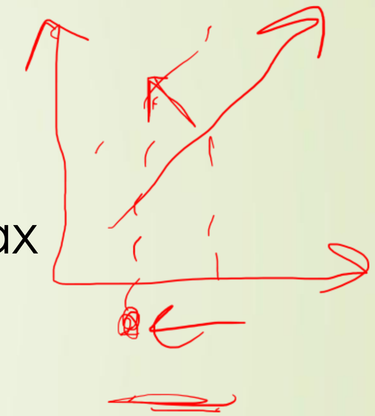
E.g.: Government imposes a $2 tax on tobacco products



E.g.: Government imposes a sales tax of 20%

How taxes on Sellers affect Market Outcomes (Cont'd.)

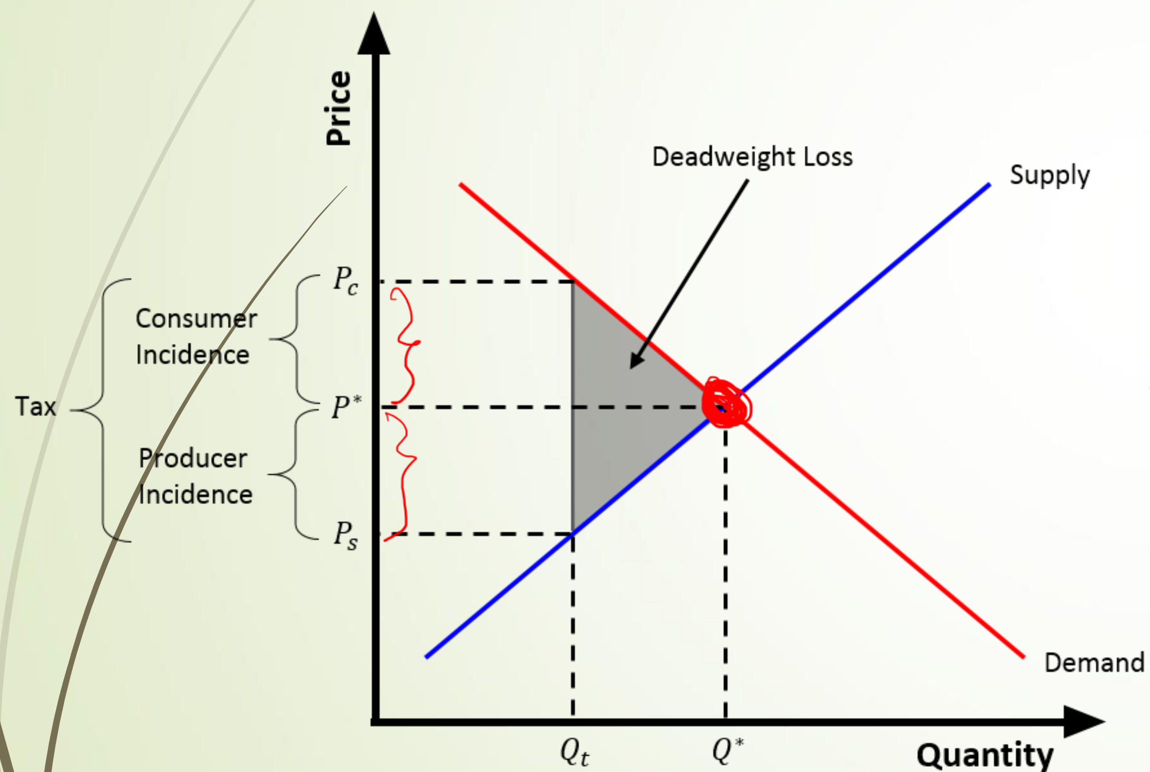
- When we consider the immediate impact of the tax, as a result of tax, cost of production goes up and producers (suppliers) will not enjoy the same profit. Thus, they will not supply the same to the market.
- Hence, supply will be reduced (an upward shift).
- Economists use the word **tax incidence** to refer to the distribution of a tax burden.



- This will be determined by the relative elasticity of demand and supply.
- If the elasticity of demand is lesser than the elasticity of supply ⇒ greater potion will be forwarded to consumer.

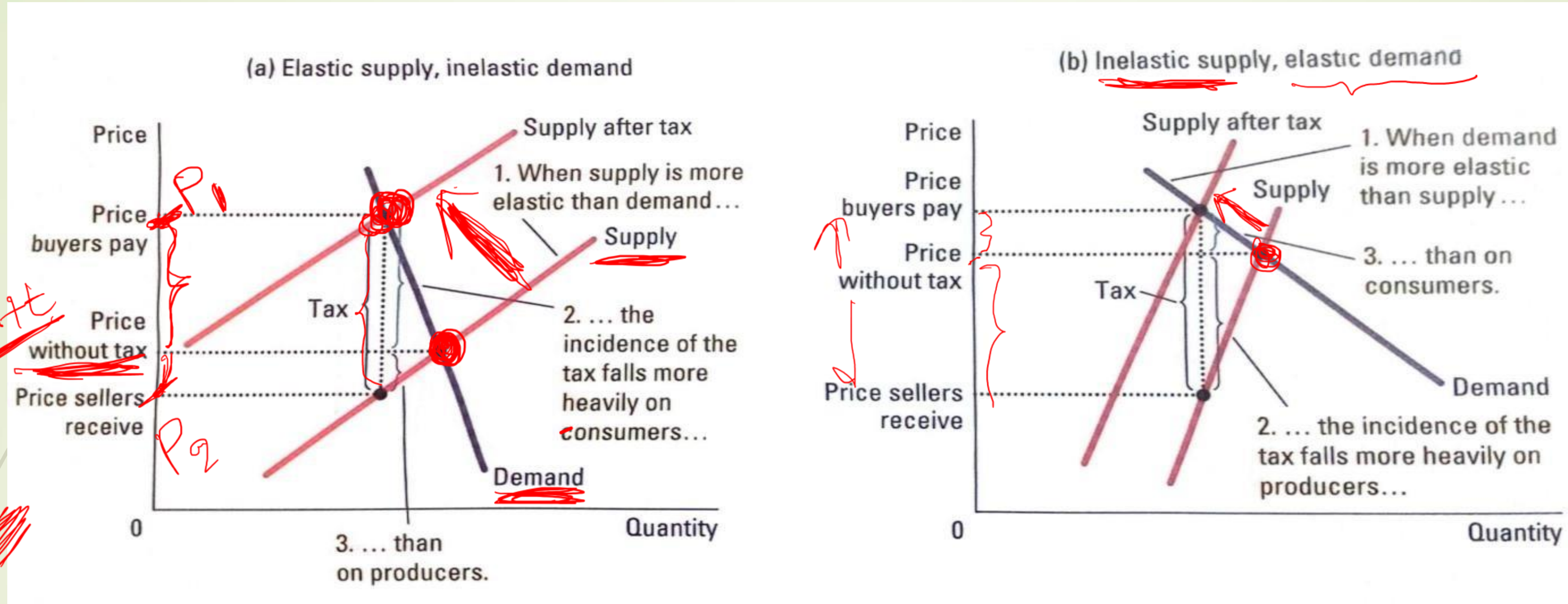
Incidence of tax

Tax incidence is an economic term that describes the final resting place of a tax burden—specifically, who actually ends up paying the tax. While a law might dictate that a seller must pay a tax, the seller may pass that cost on to the consumer through higher prices, shifting the "incidence" to the buyer.



- New price level is not equal to P* + tax since the entire tax is not passed on to consumer.
- The price will only rise up to Pc and Pc-P* is the amount forwarded to consumer.
- Ps is the amount that suppliers actually get because the tax amount should actually be paid to the government by the producer and will not be able to pass it on entirely to consumers.
- P*– Ps is the amount absorbed by the producer.

Elasticity of supply and incidence of tax



- **The Outcome:** When the tax is imposed, the "Supply after tax" curve shifts up. Because consumers are unwilling or unable to significantly reduce their quantity demanded (perhaps it's a necessity), they end up absorbing most of the price increase.

- **Tax Burden:** The vertical distance from the "Price without tax" up to the "Price buyers pay" is much larger than the distance down to the "Price sellers receive."

- **Key Takeaway:** The incidence of the tax falls **more heavily on consumers**

- **The Outcome:** If the price for consumers rises too much, they will stop buying the product. To keep the market moving, sellers have to "swallow" most of the tax costs themselves.

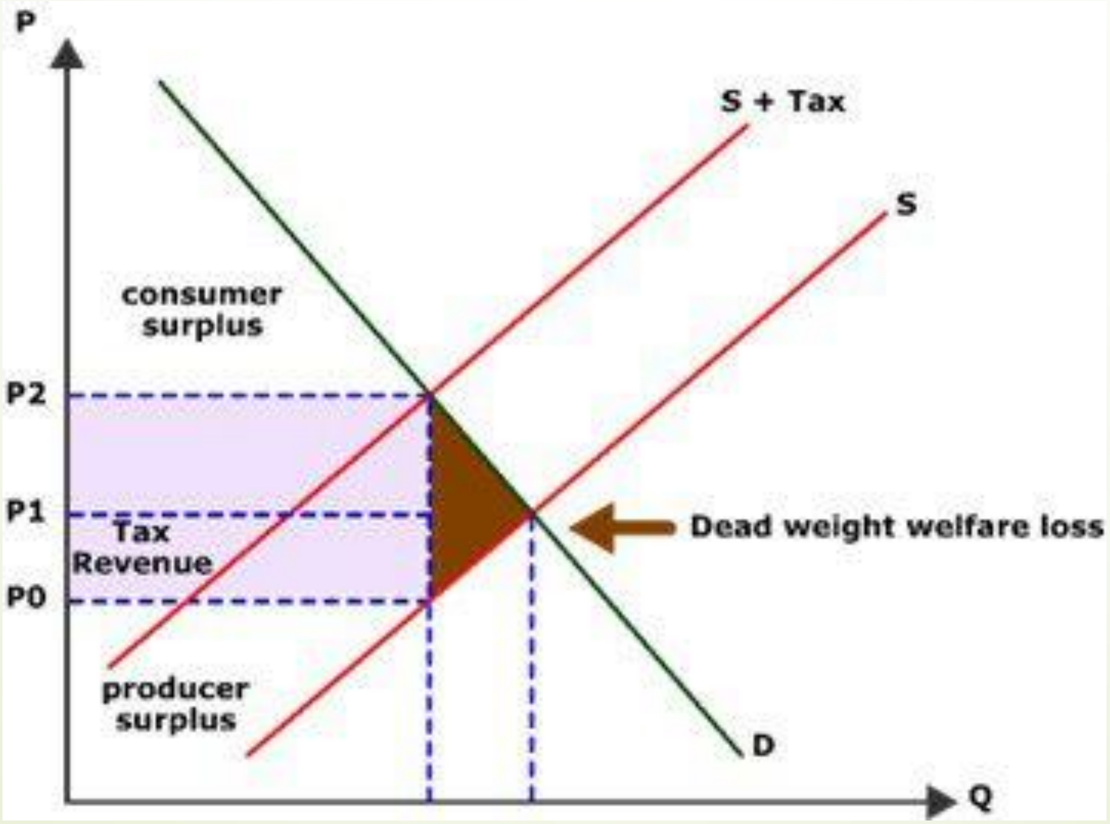
- **Tax Burden:** The vertical distance from the "Price without tax" down to the "Price sellers receive" is much larger than the increase felt by the buyers.

- **Key Takeaway:** The incidence of the tax falls **more heavily on producers**

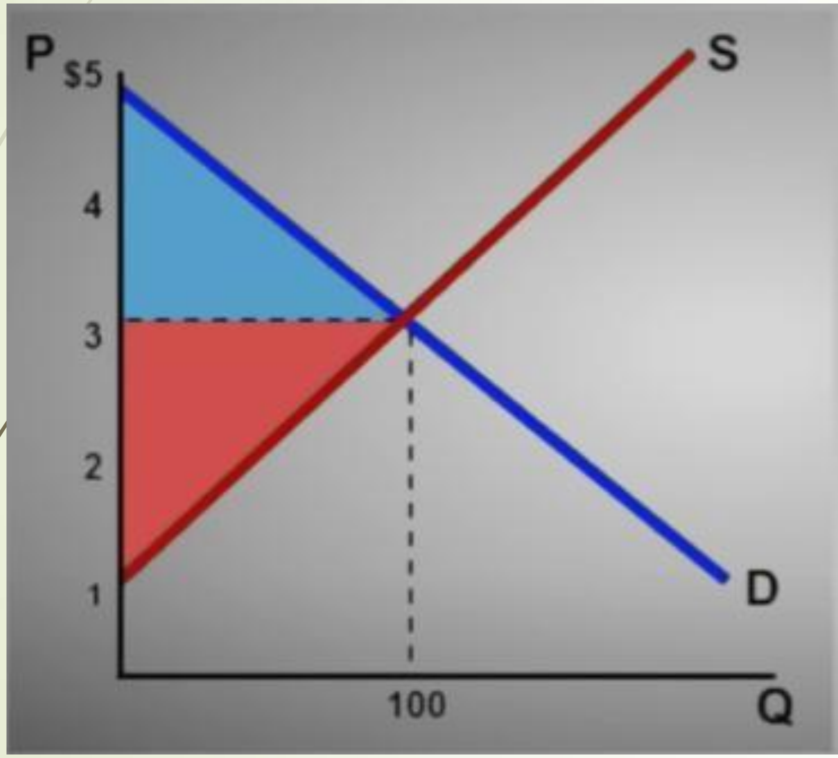
The deadweight loss of tax (DWL)

Deadweight loss (DWL) of a tax refers to the **loss of total economic welfare** (consumer + producer surplus) that occurs because a tax distorts market behavior. It represents **trades that no longer happen** due to the tax.

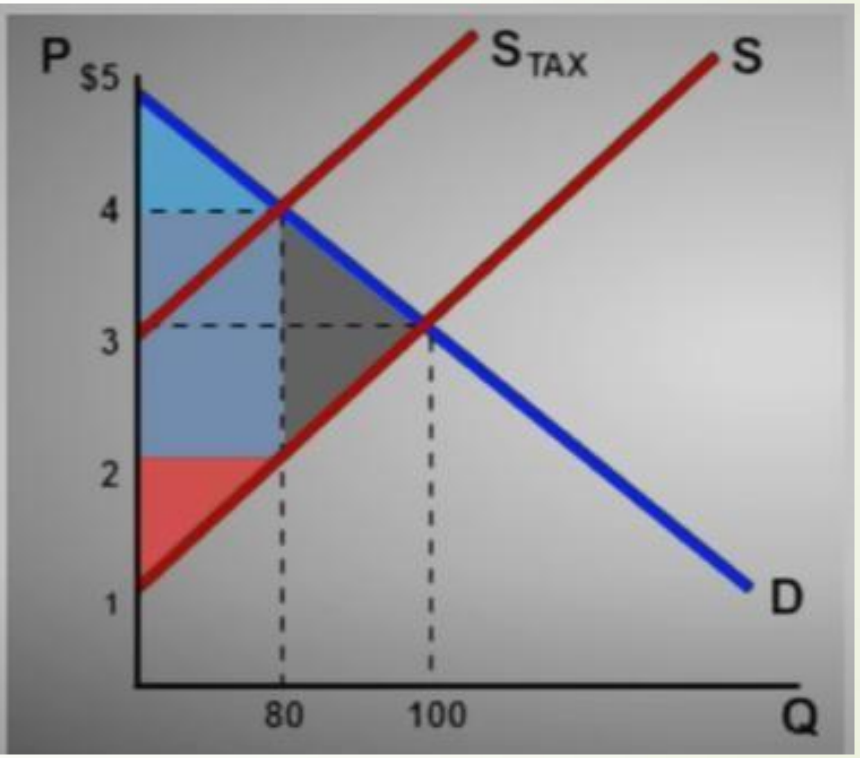
In simpler words: when a tax is imposed, some buyers and sellers who would have voluntarily traded at the original price stop trading because the price has increased for buyers or decreased for sellers. The value of these "lost trades" is the DWL.



Before Tax



After Tax



Subsidies

Government provide subsidies to encourage production.

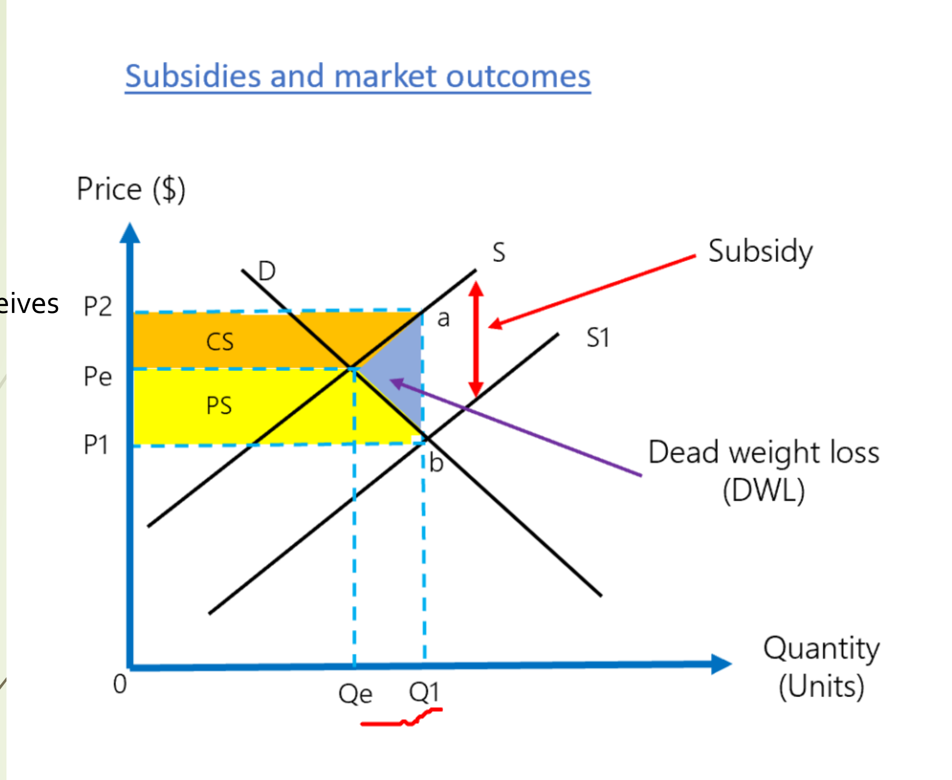
Producers will enjoy higher profits at the same price level. Hence, they increase supply.

There will be downward shift of the supply curve.

How much of will be enjoyed by supplier will be determined by the relative elasticity of demand and supply.

Subsidies

Producer receives this price



P1 (The Price Buyers Pay):

P2 (The Price Sellers Receive):

P2 = P1 + Subsidy per unit.

This graph illustrates the economic impact of a **government subsidy** on a market. A subsidy is essentially the opposite of a tax—it is a payment from the government to producers to encourage them to supply more of a good or service.

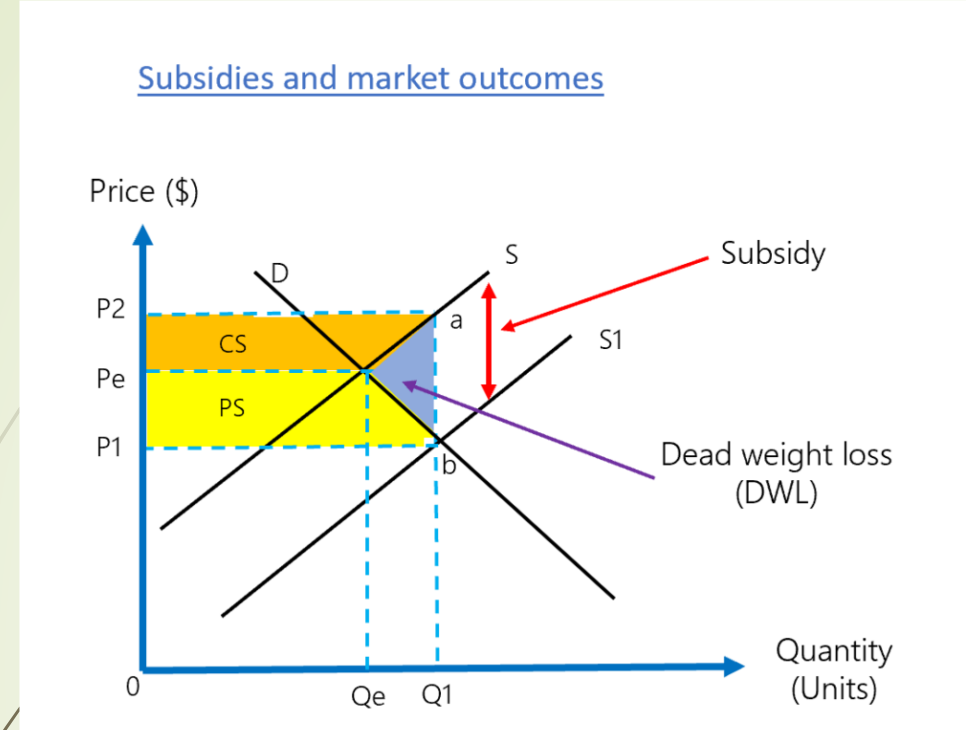
Key Components of the Graph

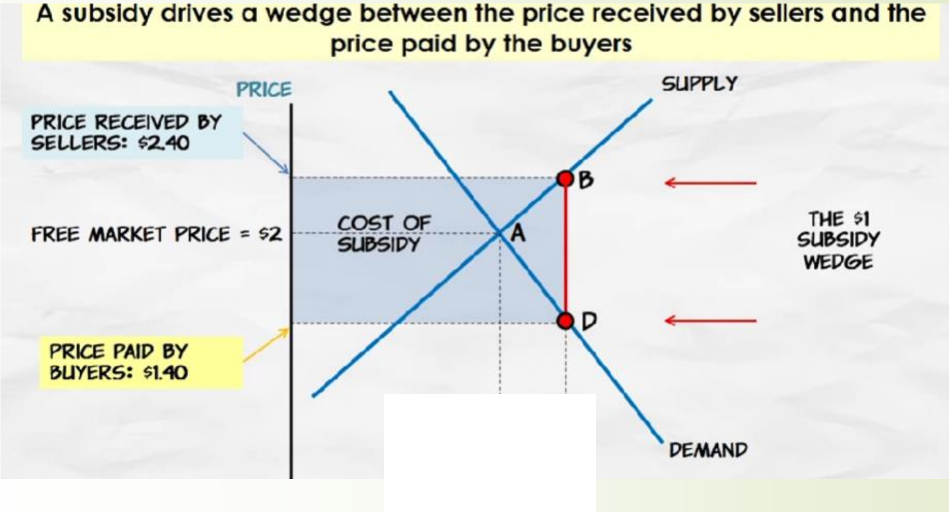
- **Shifting Supply:** The supply curve shifts right (from S to S1) because the subsidy lowers the cost of production for sellers, allowing them to provide more units at any given price.

- **New Equilibrium:** The market moves from the original equilibrium (Pe, Qe) to a new point where the quantity increases to $Q1$.

- **Price Wedge:** The subsidy creates a gap. Producers receive price P2 (the market price plus the subsidy), while consumers pay a lower price, $P1$.

Subsidies





This policy creates several important economic shifts:

1. **Increased Surplus:** Because the price for consumers falls (to P1) and the quantity increases, both **Consumer Surplus (CS)** and **Producer Surplus (PS)** expand relative to the original equilibrium.
2. **Deadweight Loss (DWL):** This is the crucial takeaway. Even though total surplus might seem higher, the economy suffers a "loss" represented by the blue triangle.
   - This happens because the government is spending money to produce units between Qe and Q1 where the **cost of production** (shown on the supply curve S) is actually higher than the **value to consumers** (shown on the demand curve D).
   - In economic terms, we are "over-producing" beyond the efficient level, which results in an inefficient allocation of resources.

Next.....

- ✓ **Theory of Production**